

EL&C ZFH4 - Partial Derivatives

- learning because i thought i knew how...
↳ i gaslit myself & doomed myself for the midterm

Youtube: "Partial Derivatives - Multivariable Calculus" → Organic Chem Tutor

- $f(x, y) = 7x^2 - x^3y^4 + 5x^4y^3$

$$f_x = 14x - 3x^2y^4 + 20x^3y^3 \quad \checkmark$$

$$f_y = 0 - 4x^3y^3 + 15x^4y^2 \quad \checkmark$$

- $z = 3x^2y^4 - 5x^7 + 4y^8 = f(x, y)$

$$f_x = 6xy^4 - 35x^6 + 0 \quad \checkmark$$

$$f_y = 12x^2y^3 - 0 + 32y^7 \quad \checkmark$$

- $z = e^{x^2y^3}, \quad \frac{d}{du}[e^u] = e^u \cdot u'$

$$\frac{dz}{dx} = 2xy^3 \cdot e^{x^2y^3} \quad \checkmark$$

$$\frac{dz}{dy} = 3x^2y^2 \cdot e^{x^2y^3} \quad \checkmark$$

- $z = x^4 e^{y^5}$

$$\frac{dz}{dx} = 4x^3 \cdot e^{y^5} \quad \checkmark$$

$$\frac{dz}{dy} = x^4 \cdot 5y^4 e^{y^5} \quad \checkmark$$

- $f(x, y) = \ln(x^2 + y^2), \quad \frac{d}{du}[\ln(u)] = \frac{u'}{u}$

$$\frac{df}{dx} = \frac{2x+0}{x^2+y^2} \quad \checkmark$$

$$\frac{df}{dy} = \frac{0+2y}{x^2+y^2} \quad \checkmark$$

- $z = \ln\left(\frac{x^2}{y}\right) = \ln(x^2 y^{-1})$

$$\frac{dz}{dx} = \frac{2x y^{-1}}{x^2 y^{-1}} = \frac{2x}{x^2} = \frac{2}{x} \quad \checkmark$$

$$\frac{dz}{dy} = \frac{x^2 y^{-2}}{x^2 y^{-1}} = \frac{-y^{-2}}{y^{-1}} = -y^{-1} = -\frac{1}{y} \quad \checkmark$$

$$\circ f(x, y) = \sqrt{x^2 + y^2} = (x^2 + y^2)^{1/2}$$

$$\frac{df}{dx} = \cancel{\frac{1}{2}} (x^2 + y^2)^{-1/2} \cdot (\cancel{2}x + 0) = \frac{x}{\sqrt{x^2 + y^2}} \quad \checkmark$$

$$\frac{df}{dy} = \cancel{\frac{1}{2}} (x^2 + y^2)^{-1/2} \cdot (0 + \cancel{2}y) = \frac{y}{\sqrt{x^2 + y^2}} \quad \checkmark$$

$$\circ z = \sin(x^3 y^5), \quad \frac{d}{du} [\sin(u)] = \cos(u) \cdot u'$$

$$\frac{dz}{dx} = \cos(x^3 y^5) \cdot 3x^2 y^5 \quad \checkmark$$

$$\frac{dz}{dy} = \cos(x^3 y^5) \cdot 5x^3 y^4 \quad \checkmark$$

• Evaluate f_x & f_y @ $(1, 2)$

$$\circ f(x, y) = 2x^3 y^2 + 5y^3 + 4x^2$$

$$\frac{df}{dx} = 6x^2 y^2 + 0 + 8x$$

$$@ (1, 2) = 6(1)^2 (2)^2 + 0 + 8(1)$$

$$= \underbrace{\hspace{10em}}$$

$$\frac{df}{dy} = 4x^3 y + 15y^2 + 0$$

$$@ (1, 2) = 4(1)^3 (2) + 15(2)^2$$

$$= \underbrace{\hspace{10em}}$$